ARXIS - Traceability does not grant approval

Private portfolio design study. Not a client deployment.

Problem and decision user

A changed requirement can leave inconsistent decisions, versions and deliverables in circulation. The proposed user is design lead, documentation coordinator and project reviewer.

Operating and economic approach

An operating model for traceable design documentation and controlled project change. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Seven tests check every decision-to-artifact obligation, duplicates, unknown references and version/actor controls. A matching version permits human review, not automatic engineering approval.

Capabilities illustrated

Requirements; document governance; change control; cross-functional coordination. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Traceability completeness does not prove correctness of engineering design or compliance with building law. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study for one synthetic residential project. The product is a controlled documentation workflow, not a CAD/BIM/GIS integration or an engineering approval service.

Problem, customer and alternatives

A project manager needs to know which requirement caused a decision, which artifact implements it, who may approve a change and which downstream work must be reconsidered. A folder of drawings alone does not answer that question. The proposed users are the project manager, discipline leads, document controller and client representative. These are role designs, not a staffed organization.

Alternatives are a controlled spreadsheet/register, an existing common-data environment, or a custom traceability layer. Prefer an exportable existing tool unless a concrete cross-document dependency cannot be handled. No paid demand, procurement outcome or reduction in construction errors is established.

Information contract

Requirement: ID, statement, source, owner, acceptance criterion and status. Decision: ID, requirement links, alternatives, rationale, approver, date and effective version. Artifact: ID, decision links, version, author, checker, approval state, confidentiality and content hash. Change request: affected requirement, reason, expected source versions, cost/time estimate, reviewer, authority and disposition.

The synthetic fixture names three illustrative documents; these names are not actual architectural drawings or certified engineering reports. Rule-based document classification is only a suggestion requiring review; it is not a trained AI classifier or technical compliance check.

Operating workflow

1. Project manager captures the client need and mandate. Discipline lead translates it into reviewable requirements; document controller registers IDs and versions.
2. Analyst prepares alternatives. The authorized decision owner records the selected option, unresolved assumptions and acceptance criteria.
3. Author produces an artifact. Checker confirms every linked decision is addressed; one artifact somewhere under a requirement is not sufficient if another decision remains uncovered.
4. A change request identifies the expected versions before review. The reference rejects unknown actors, flags version mismatch and returns affected decisions/artifacts. It never grants engineering approval.
5. Discipline leads assess cost, schedule, constructability and adjacent requirements. Project manager consolidates dependencies. The authorized owner approves, rejects, pauses or requests redesign.
6. Document controller issues the new baseline, preserves superseded versions and informs affected recipients. Unreviewed downstream documents become stale, not silently current.

Economics and constraints

The synthetic management case assumes four coordination hours at EUR 60 plus EUR 10 allocated tooling cost: EUR 250 per change. Eight hours cost EUR 490. These are coordination costs, not project savings or total construction change costs. Materials, contractor variations, delay penalties and design liability are excluded.

A change memo must separately show direct implementation cost, review effort, schedule impact and uncertainty. Missing cost or date information pauses a definitive recommendation. Compare improved traceability against its data-entry and governance burden; extra fields are justified only when they change a decision or prevent a specific error.

Architecture and tool boundaries

Canonical requirements and decisions remain in exportable records; diagrams and task boards are views. A document platform must support immutable issued versions, access control and change history. The local reference checks graph structure and explicit version/actor inputs only. An approver name passed to a function is not authentication or secure authorization. Real deployment requires identity integration, immutable audit logging and specialist review.

Use the course-derived iterative requirements approach to clarify user stories, update the backlog and review a changed need with stakeholders. This system remains event-driven; the course's sprint cadence is not an imposed calendar requirement. Existing architecture/construction materials provide candidate domain extensions, not permission to claim professional certification or code compliance.

Dashboard, risks and acceptance

Metrics: requirements with all decisions represented; orphan/dangling links; stale artifacts; pending approvals by owner; coordination cost and age of unresolved changes. Every denominator uses a declared project baseline. Completeness is a structural property, not proof that the design is safe or correct.

Negative cases: duplicate IDs, unknown references, an uncovered second decision, unauthorized actor, old expected version and missing artifact. Hand off the fixture, dependency matrix, change assessment, limitations and review memo. Pause when authority or baseline is unclear. Stop a custom-tool build if an existing controlled register meets the same needs more cheaply.

Interview narrative: the main contribution is separating traceability from approval. Review exposed a false-completeness case where one artifact hid another uncovered decision. Corrected checks and version-aware impact assessment make the governance question explicit without claiming engineering validation.

Validation evidence and reflection

Private synthetic study. Seven repository tests check traceability, missing artifacts, duplicate/dangling IDs, uncovered decisions, rule-based classification disclosure and actor/version controls.

Result and forecast

The complete three-requirement fixture has structural coverage; removing an artifact or adding an uncovered decision exposes a gap. A version mismatch pauses the change; a recognized actor with matching versions is ready for human impact review, never automatically approved. The original completeness/negative-case prediction holds within this authored fixture.

The pre-recorded coordination-cost case gives EUR 250 per change at four hours and EUR 490 at eight hours. This is assumed effort cost, not a measured saving. Doubling review time does not double the fixed EUR 10 tooling allocation.

Decision and limitations

Continue with a controlled requirements/decision/version register and a human change-review workflow. Do not infer engineering correctness from graph completeness or deploy the actor-input check as an access-control system. Obtain actual scope, approver mandate, cost and schedule implications before issuing a change.

Post-mortem

Earlier completeness checked whether any artifact existed beneath a requirement; it could conceal an uncovered second decision. The revised check requires every linked decision to be represented. Duplicate and dangling references are rejected. The reference classifier remains deterministic keyword matching, not a trained AI model. A controlled spreadsheet or existing document system may meet the need without new software.

Handoff: DELIVERY-SPECIFICATION.md, source fixture, tests and artifacts/management-review.json. No client adoption, construction saving or regulatory compliance is claimed.